

Everything the camera cannot measure about the shuttle, the floor can

a broadcast: track, players



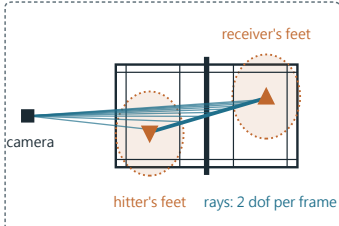
one fixed camera, no calibration, no 3D truth

b camera from lines + cord



court model through the fitted camera: lines 1.4 px, cord 1.7 px

c rays + planar anchors



d tier 1: Shuttle3D (depth prior)

2.7 M parameters; synthetic flights only; never a pixel; milliseconds per stroke

head: one depth r per ray; $X = C + r \cdot d_0 \text{ ray}$



Transformer encoder (6 layers, width 192)
global token added to every ray token

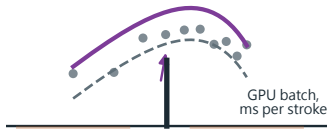


ray tokens (one per frame)

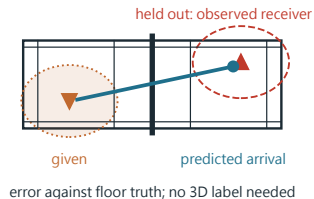
global token: camera, feet, receiver, landing, wrist z (grey = absent)

e tier 2: rectifier

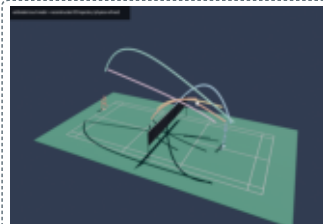
grey: per-frame depths; dashed: launch-state fit
purple: projected onto rules + anchor discs



f leave-one-anchor-out

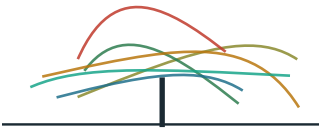


g output: flight + players in 3D



h synthetic supervision (trains d)

13,010 real strokes: start, end and flight time
aerodynamic two-point problems: 77,728 flights
32 measured cameras, 7 venues held out; audited



— solid border: trainable, back-propagated (only d)

- - - - - dashed border: analytic / physics solver, no learned weights

a possible flight in the court frame, or the stroke flagged unreliable